

# Full Spectrum Nonperturbative Higgs Vacuum Decay via Lattice Euclidean Methods in 4D

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## Project Description

This project aims to perform the first nonperturbative, lattice–Euclidean computation of the electroweak vacuum decay rate, including the full fluctuation–operator spectrum and determinant ratio around the Coleman bounce. Although the semiclassical bounce formalism has existed since the late 1970s, the prefactor governing false–vacuum decay has never been computed from first principles in four dimensions. The proposed work adapts proven lattice–QCD spectral–extraction techniques to solve this long–standing problem.

The stability of the Higgs vacuum is a central question in particle physics and cosmology. Current estimates rely entirely on perturbative renormalization–group extrapolations and one–loop semiclassical approximations. These methods cannot capture nonperturbative effects, finite–volume corrections, multi–field dynamics, or early–universe backgrounds. A lattice–Euclidean approach provides controlled errors, systematic improvement, and ab initio access to the fluctuation spectrum that determines the decay prefactor.

The false–vacuum decay community has historically focused on analytic semiclassical methods, while the lattice community has focused almost exclusively on QCD. The determinant–ratio problem sits at the intersection of these fields, requiring both bounce–physics expertise and lattice spectral–extraction experience. The PI’s prior work provides this missing bridge.

Twenty years ago, the PI developed a lattice–QCD spectral–extraction framework for hybrid–meson decay widths, published in *Physical Review D*. That work introduced a computational pipeline—operator construction, correlation–matrix formation, variational diagonalization, finite–volume spectral mapping, and width extraction. The mathematical structure of false–vacuum decay is directly analogous: the bounce replaces the hadronic operator; the fluctuation operator replaces the correlation matrix; eigenvalue extraction proceeds identically; and finite–volume mapping parallels Lüscher–type methods.

Initially, the methodology from the previous decay–width calculation will be mapped onto false–vacuum field decay as described on the next page. The approach will first be tested using MATLAB and/or Mathematica before being transferred to a high–performance computing center for four–dimensional calculations. For cost–accounting purposes, this work will be conducted under the auspices of *In Vero Vacuo, LLC* according to the four–stage plan described on page 3.

## References

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# Mapping of Decay Width Lattice QCD onto Higgs Field False Vacuum Decay

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## Operator Construction (Hybrid Mesons)

Build operators with correct  $J^{PC}$ ; include gluonic excitation structure; symmetry projections.

### ⇒ Bounce Construction (Vacuum Decay)

Build  $O(4)$ -symmetric bounce; enforce boundary conditions; stabilize Euclidean solution.

## Correlation Matrix Formation

Construct  $C_{ij}(t)$ ; Hermitian matrix encodes spectral content; contraction and noise control.

### ⇒ Fluctuation Operator

Construct Hessian  $H = \delta^2 S / \delta \phi^2$ ; large sparse operator; full fluctuation spectrum.

## Variational Diagonalization

Solve  $C(t)v = \lambda(t)C(t_0)v$ ; extract excited states; stabilize eigenvectors.

### ⇒ Eigenmode Extraction

Solve  $H\psi_n = \lambda_n\psi_n$ ; identify negative mode, zero mode, positive spectrum.

## Smearing Techniques

Improve operator overlap; reduce UV noise; enhance signal quality.

### ⇒ Bounce Stabilization

Smooth bounce profile; condition fluctuation operator; stabilize negative/zero modes.

## Effective Mass Analysis

Extract masses from  $m_{\text{eff}}(t)$ ; identify plateaus.

### ⇒ Mode-Decay Analysis

Extract Euclidean frequencies; validate negative and translational zero modes.

## Lüscher Finite-Volume Mapping

Relate discrete spectra to continuum phase shifts.

### ⇒ Determinant-Ratio Mapping

Compute  $\det'(H)/\det(H_0)$ ; remove zero mode analytically; finite-volume scaling.

## Breit-Wigner Width Extraction

Fit phase shifts; obtain resonance width.

### ⇒ Full Vacuum-Decay Rate

Compute prefactor  $A = (\det' H / \det H_0)^{-1/2}$ ; combine with bounce action  $S_B$ .

## Mapping Summary

|                       |                        |
|-----------------------|------------------------|
| Operator Construction | ⇒ Bounce Construction  |
| Correlation Matrix    | ⇒ Fluctuation Operator |
| Diagonalization       | ⇒ Eigenmode Extraction |
| Smearing              | ⇒ Bounce Stabilization |
| Effective Masses      | ⇒ Mode Frequencies     |
| Lüscher Mapping       | ⇒ Determinant Ratio    |
| Breit-Wigner Width    | ⇒ Full Decay Rate      |

# Four-Stage Project Plan

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## Stage 1: Foundational Development

- Reproduce benchmark  $O(4)$  bounce solutions in MATLAB and/or Mathematica and establish a documented computational workflow.
- Construct small-lattice discretizations of the Euclidean action and fluctuation Hessian; identify the negative mode, translational zero modes, and the lowest positive modes.
- Compare the low-lying spectrum and determinant ratio with known analytic or semiclassical limits, delivering prototype code and a quantitative feasibility assessment.

## Stage 2: Full Lattice Implementation

- Transfer the validated algorithms to a high-performance computing environment with scalable sparse-matrix construction and eigensolvers.
- Extract approximately 50–200 low modes, estimate the remaining spectral contribution, and implement the analytic removal and normalization of zero modes.
- Perform lattice-spacing and finite-volume studies, construct an uncertainty budget, and validate the result against the weak-coupling Coleman–Callan prediction.

## Stage 3: Standard Model Higgs Potential

- Incorporate the renormalization-group-improved Higgs potential and relevant electroweak input parameters into the lattice framework.
- Compute the bounce, fluctuation spectrum, determinant prefactor, and decay rate while tracking discretization, finite-volume, and parameter uncertainties.
- Compare the nonperturbative result with conventional perturbative lifetime estimates and identify the parameters that dominate the metastability uncertainty.

## Stage 4: Multi-Field and Cosmological Extensions

- Extend the method to coupled scalar fields, including Higgs-portal and higher-dimensional effective interactions.
- Adapt the calculation to finite temperature, inflationary or curved backgrounds, and the quantum-to-thermal decay crossover.
- Produce reusable computational tools and metastability maps showing how new fields and cosmological conditions modify the decay rate.